Hybrid Tensor-Train Time Evolution: Status, Findings, and Outlook

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TN_Turbulence project — 2D passive scalar study
(Dated: October 18, 2026)

Concise status note on the machinery built to evolve a 2D scalar field in the hybrid tensor-train (TT) encoding of the compression study. Records the equations being solved, what has been implemented, what has been checked, the findings and caveats so far, and the plan going forward. Detailed numbers live in the test suite; this note keeps only the conclusions. As of this revision the machinery is validated, a genuine cascade has been run, and the central question — whether the TT rank stays bounded once fine scales are generated — has been answered affirmatively.

I. THE EQUATIONS

A. Passive scalar advection–diffusion

The target problem, and the one the compression study characterises statistically:

$$\partial_t\theta + \mathbf{u}\cdot\nabla\theta \;=\; \kappa\nabla^2\theta + f, \qquad \nabla\cdot\mathbf{u} = 0, \tag{1}$$

on a doubly periodic square. Here θ is a scalar (temperature, dye concentration) carried passively by a prescribed incompressible velocity $\mathbf{u}$: it does not react back on the flow. κ is the molecular diffusivity and f a large-scale source.

a. Physics. Advection stretches and folds θ, cascading scalar variance from the injection scale down to the Batchelor scale where diffusion erases it. The resulting fields are strongly *intermittent*: increments have heavy tails and flatness grows without bound as the separation shrinks, which is precisely the property the hybrid encoding was designed to preserve. The variance budget follows from Eq. (1),

$$\frac{d}{dt}\,\tfrac{1}{2}\langle\theta^2\rangle \;=\; -\kappa\big\langle|\nabla\theta|^2\big\rangle \;+\; \langle\theta f\rangle, \tag{2}$$

because the advective term is skew-symmetric for $\nabla\cdot\mathbf{u} = 0$ and therefore conserves $\langle\theta^2\rangle$ exactly. Note the consequence, which matters for scheme design: the *nonlinear* (bilinear) transport term is the conservative one, and the *linear* diffusive term is what dissipates. The same structure holds for Navier–Stokes, where $\mathbf{u}\cdot\nabla\mathbf{u}$ conserves kinetic energy and viscosity dissipates it.

When $\nabla\cdot\mathbf{u} = 0$ the transport term also admits the *conservative* (flux) form

$$\mathbf{u}\cdot\nabla\theta \;=\; \nabla\cdot(\mathbf{u}\,\theta), \tag{3}$$

which is equivalent analytically but not after discretisation: the flux form respects the variance budget far better and is the usual choice.

B. Cases solved so far

Work to date has deliberately used constant-coefficient reductions of Eq. (1), because they have exactly known solutions and therefore isolate compression error from every other error source:

$$\partial_t\theta = \kappa\nabla^2\theta \qquad \text{pure diffusion} \tag{4}$$

$$\partial_t\theta + \mathbf{c}\cdot\nabla\theta = 0 \qquad \text{rigid translation, } \mathbf{c} \text{ constant} \tag{5}$$

$$\partial_t\theta + \mathbf{c}\cdot\nabla\theta = \kappa\nabla^2\theta \qquad \text{constant-velocity advection–diffusion} \tag{6}$$

A genuine, space-dependent velocity has since been added: the steady cellular flow obtained from the streamfunction $\psi = \sin x \sin y$, i.e. $\mathbf{u} = (\sin x\cos y,\ -\cos x\sin y)$, run both decaying and weakly forced. The simpler choice $\mathbf{u} = (\sin y,\ \sin x)$ was tried first and discarded: it is incompressible but degenerate, in that the advective and conservative discretisations coincide identically, so it cannot distinguish the transport forms at all.

A frozen DNS velocity snapshot has since been added as a third case. The streamfunction is recovered from the stored v_x, v_y at 4096^2 by inverting the vorticity spectrally, coarsened to the target grid, and differentiated with the

solver's own MPOs. Its rank behaviour differs sharply from the cellular flow: at $L = 9$ and $\epsilon = 10^{-6}$ the streamfunction saturates the geometric bond bound and its tensor train costs more parameters than the dense field ($\rho = 1.33$, and 2.32 at $L = 10$). It becomes compressible only at much looser tolerance, $\epsilon = 10^{-2}$ giving bond 21 at $\rho = 0.084$ for a 1.7% error in ψ. Runs with the DNS velocity therefore use a truncated velocity field, which is a modelling choice recorded in each run file; the derived velocity carries bond 34 against 4 for the cellular flow.

The velocity is built from ψ with the solver's own stencils, $\mathbf{u} = (\mathsf{D}_y\psi,\ -\mathsf{D}_x\psi)$, which makes it divergence-free to round-off at the *discrete* level since D_x and D_y act on different axes and commute exactly. (For separable trigonometric flows analytic sampling happens to be discretely divergence-free too; the streamfunction route is the general guarantee, not an improvement in that special case.)

C. Standard ways to solve these

- **Pseudo-spectral** is the reference method for periodic 2D turbulence: derivatives are exact in Fourier space, the nonlinear product is formed in physical space with 2/3 dealiasing, and diffusion is handled by an integrating factor or exponential time differencing so it imposes no timestep restriction. This is how the DNS database underlying the paper was produced.
- **Finite difference / finite volume** with upwind or WENO reconstruction for the transport term, central stencils for diffusion, and SSP Runge–Kutta in time. Cheaper per degree of freedom, more dissipative, and much easier to express as a local operator.
- **Stiffness.** Explicit diffusion requires $\Delta t \le \Delta x^2/4\kappa$ in 2D, which becomes punishing on fine grids; implicit or exponential treatment of the diffusive term is standard.

a. What we do instead. The field is stored as a hybrid TT over the interleaved (Morton) bit ordering, and every spatial operator is a matrix-product operator acting on it. Stencils are finite differences, so the semi-discrete system is

$$\frac{d\boldsymbol{\theta}}{dt} \;=\; \mathsf{A}\,\boldsymbol{\theta}, \qquad \mathsf{A} \;=\; -c_x\mathsf{D}_x - c_y\mathsf{D}_y + \kappa\,\mathsf{L}, \tag{7}$$

with A collapsing to a *single* low-bond MPO in the constant-coefficient case. Time stepping is explicit Runge–Kutta, and the state is recompressed after every stage. Compression is the entire source of novelty and the entire source of risk.

II. THE ENCODING

Writing the hybrid ansatz out,

$$\theta(x,y) \;=\; \sum_{\alpha=1}^{\chi} A^0[B,\alpha]\;\Phi_\alpha(\text{fine coordinates}), \tag{8}$$

with B the coarse-block label carried by the dense head and Φ_α the fine-scale shape functions carried by the compressed tail. So the hybrid TT is a **block-wise reduced basis with a globally shared dictionary**. Two consequences:

- The coarse-grained field is a linear functional of the head alone, so "the large scales live in the head" is literally true.
- The large scales carry no rank error. For advection-dominated dynamics this is the central argument for the encoding: exact large-scale transport with rank-compressed small-scale structure.
- The same framing predicts the *seam*: neighbouring blocks get independent amplitude vectors and nothing enforces continuity across a block boundary.

a. Conventions. L = bits per axis (grid $2^L \times 2^L$); $2L$ total interleaved legs, even legs carrying rows (y) and odd legs columns (x); N = legs held by the dense head, so the head spans $2^{N/2} \times 2^{N/2}$ coarse blocks and the seam period is $2^{L-N/2}$ — *not* $2^{N/2}$, which coincides only at $N = L$.

III. WHAT WAS BUILT

All of it sits on top of `tensnet` via its public API. No `tensnet` source is modified, and — deliberately stricter — no `tensnet` global state is read or written, since a wrong global would not raise but would silently produce wrong physics.

- `hybrid_ops.py` — hybrid TT algebra: construction and contraction, truncated SVD with a genuine amplitude tolerance, two-sweep compression with *per-bond* caps, exact arithmetic (add / scale / axpy / Hadamard), MPO application by un-fusing the head, application through an explicit fused $2^N \times 2^N$ head operator, rank-1 mass projection, and pointwise evaluation without contracting the TT.
- `hybrid_operators.py` — named stencils, derivative and Laplacian MPOs, exact translation, the combined advection–diffusion operator, exact MPO algebra, and the isotropic seam smoother.
- `hybrid_evolve.py` — Euler / RK2 / SSPRK3 / RK4, a truncation policy with optional loss accounting, a stable-timestep helper enforcing both the advective and diffusive limits, variable-coefficient transport in advective / conservative / skew form with optional forcing, velocity from a streamfunction, dense finite-difference and analytic spectral references, and diagnostics including a variance-budget measure of numerical dissipation.
- Four test suites, ∼117 assertions, all against dense references.

All of this lives in `time_evolution/`, deliberately separated from the paper material in the repository root. The dependency runs one way only: these modules read `config.py` and `hybrid_MPS.py` from the root, and nothing in the root imports from them, so the paper figures reproduce without this directory.

The seam smoother is the isotropic repair — the `_int64_xy` treatment, not the x-only one. With $P_{x,y}$ the seam projectors, $Q = I - P$, and $A_{x,y}$ the periodic two-neighbour averages,

$$S \;=\; Q_x Q_y \;+\; P_x Q_y A_x \;+\; Q_x P_y A_y \;+\; P_x P_y \tfrac{1}{2}(A_x + A_y). \tag{9}$$

IV. WHAT HAS BEEN CHECKED

Everything below is verified numerically against a dense reference, not asserted from reading source.

- Lossless hybrid round trip; agreement between our constructor and the legacy one; the index layout and seam geometry.
- All arithmetic is exact, with no hidden truncation.
- Compression: lossless at full rank, monotone in the cap, head physical dimension preserved, head bond capped independently of the tail bonds, and norm genuinely lost when truncating.
- Every operator — first and second derivatives at 2nd and 4th order, upwind stencils, Laplacian, translation, seam smoother — against dense, on both axes, at several (L, N).
- Operator composition, and agreement between composing then applying versus applying twice.
- The seam smoother reproduces the dense reference implementation exactly, is isotropic under transposition, and touches only seam lines.
- All four time steppers reproduce their dense counterparts.
- The fused-head and un-fused-head paths agree to round-off, both as single applications and over multi-step evolutions.
- Pointwise evaluation agrees with the dense field.
- The variable-coefficient right-hand side against dense, for all three transport forms, with and without diffusion and forcing.
- Variance conservation of the three transport forms, at several resolutions and timestep sizes.

a. Verified `tensnet` traps. None of these raise; all silently corrupt. The global 2D architecture defaults to the non-interleaved layout; `get_diff_mpo` defaults to 8th-order stencils; MPS addition and multiplication silently compress at a global l_2 tolerance; `tt.svd` treats its `eps` as a squared-energy fraction, so the achievable amplitude error saturates around $\sqrt{\epsilon}$; `tt.svd` renormalises singular values after truncation by default; `shift_mpo` has inverted flags and is non-periodic; `MPO.__matmul__` and `MPO.contract()` cannot handle hybrid shapes at all.

V. FINDINGS AND REMARKS

- **The rank saturates in a genuine cascade.** This is the central result. In the cellular flow the intrinsic rank of the *dense* solution — a property of the physics, not of the scheme — rises during the initial stretching and then plateaus for the rest of the run rather than growing without bound. An adaptive-tolerance TT tracks it a few units below throughout, with the error flat over a thousand steps. Constant-velocity transport could not test this; a real cascade can, and the encoding holds.
- **Error saturates at fixed rank too**, at every bond dimension tried, reproducing what was already seen for pure translation.

- **Bulk statistics survive rank starvation far better than pointwise accuracy.** At a third of the rank the field actually needs, the pointwise error is a few percent while the variance still tracks the reference to about one part in a thousand. Encouraging for a statistics-focused study, and worth testing directly against the paper's diagnostics.
- **Numerical dissipation is negligible against molecular dissipation** — a fraction of a percent of it, even at a third of the required rank. On this evidence plain truncation is adequate and a norm-conserving integrator is not needed, which retires the main open question about the scheme.
- **The skew-symmetric transport form is essential.** The advective and conservative discretisations drift in variance by equal and opposite amounts, and their average is conserved to machine zero — at every resolution, timestep and velocity construction tested. The drift of the one-sided forms grows sharply as the grid coarsens or the run lengthens, so it is not a small effect.
- **Centred advection constrains the time scheme.** The spatial operator has purely imaginary eigenvalues, so forward Euler is outright unstable and RK2 is *latently* unstable: its amplification factor on the imaginary axis exceeds one, which stays invisible on a smooth field and would bite precisely when the cascade generates high-wavenumber content. SSPRK3 or RK4 is required.
- **Performance came almost entirely from one change.** Compression sweeps dominate the cost, and on the matrix shapes they produce torch's CPU LAPACK path is several times slower than scipy's; switching the sweeps gave roughly an order of magnitude end to end. Thread count mattered before the switch and does not after. Two further optimisations that flop counting predicted would help — replacing an einsum with a matmul, and cutting the number of truncations per step — both measured within noise, because fewer-but-larger compressions offset the reduced call count.
- **The low-rank structure survives advection.** A one-cell translation inflates the state rank by of order ten percent, transiently, and returns it *exactly* to baseline whenever the shift is a whole number of coarse blocks. Verified at two different (L, N). An adaptive-tolerance scheme tracks the intrinsic rank of the exact solution to within one or two, so the method accumulates essentially no spurious rank.
- **Truncation error saturates rather than compounding.** Under a hard rank cap over hundreds of advection steps the error stops growing and the field settles onto a representable state that then simply translates. This was the principal risk identified at the outset and it did not materialise for constant-velocity transport.
- **Lossy truncation per step is a property of the cap, not of advection.** It appears when the rank cap sits below the field's intrinsic rank — which is exactly the fixed-compression-ratio regime the study works in, and a legitimate choice, but the loss should be attributed correctly.
- **Rank capping and tolerance capping answer different questions.** A cap fixes the compression ratio and lets error float; a tolerance fixes error and lets rank float, revealing what the physics actually demands. Both are supported and both are needed.
- **The seam smoother should not run inside the time loop.** It is a low-pass filter and is not idempotent, so iterating it is diffusion. It never beat leaving it off. The *severity* is configuration dependent — it scales with the fraction of cells on seam lines, $\approx 2^{1-L+N/2}$ — so the earlier dramatic figures came from an unrepresentatively small block width. The seam is an artifact of the representation, so its repair belongs on the output being analysed, not on the state being evolved.
- **Mass conservation is cheap and worth having.** Truncation drifts the mean; a rank-1 constant restores it to round-off at no accuracy cost, provided it is applied *after* truncation. Unlike rescaling the l_2 norm it does not move variance between scales.
- **Do not renormalise after truncation.** Truncation is an orthogonal projection and losing norm is a theorem, not a defect. Rescaling puts the discarded small-scale variance back into the retained large scales — a spurious inverse cascade — and, worse, disables the very diagnostic that detects under-resolution.
- **Consequence for the published data.** Because rescaling does not change the retained subspaces, the legacy pipeline's output is exactly plain truncation times a *global scalar* that tends to unity as the bond dimension grows. Every scale-invariant statistic — flatness, normalised structure-function ratios, normalised increment PDFs — is therefore unchanged; only the absolute level of $E(k)$ can have moved. This still needs confirming on the real DNS snapshots.
- **The fused head does not scale.** Single-stencil operators have very sparse fused heads, but operators that are sums of terms — the Laplacian, the seam smoother — have essentially dense ones, so sparsity buys nothing for exactly the operators one most wants. It is modestly faster at small N but becomes infeasible in memory at production N; the un-fusing path, whose cost carries no 4^N term, is the one that scales.
- **The cascade run was decaying, not statistically steady.** The forcing amplitude was set to balance the *initial* dissipation, but once the cascade steepens gradients the dissipation grows by orders of magnitude, so the field decays anyway. That the rank plateau holds while the variance falls by two orders is arguably a stronger

statement, but the statistically-steady state is the regime the paper's statistics describe and it has not yet been reached.

- **Traps for the experimenter.** Rank experiments need a field whose rank has room to move: a multiscale field is full rank and the measurement merely saturates the chain geometry. A total error that falls *below* the pure finite-difference floor is coincidental error cancellation, not an improvement. A diagnostic that aggregates truncation losses across a whole right-hand side does not measure dissipation of the state. And a transport test flow must be genuinely two-dimensional, or the discretisation forms it is meant to separate coincide identically.

a. The hybrid advantage survives evolution, and is a factor of about two in parameters. Benchmarked against a standard tensor train (the $N = 0$ case of the same encoder, verified to reproduce `tt.vector_to_mps` bond for bond) at four resolutions from 256^2 to 2048^2: at a fixed target accuracy the hybrid encoding needs 1.8–2.5 times fewer parameters, with no trend in resolution. The useful head sizes are $N \simeq L$, i.e. half the bits of each axis, and $N \in \{L-2, L, L+2\}$ brackets the optimum everywhere measured. Full tables are in the companion report `report_vs_standard.pdf`.

b. Representation error predicts evolution error. Across four resolutions, five head sizes and five bond caps, the error made in evolving a field lies within a factor 1.5–25 of the error of merely compressing the same field at the same budget, with most points below 5. Measuring the ordering in N and the scaling in L by compression therefore costs minutes where integration costs core-days.

c. The published compressed fields are plain truncation times a global scalar, confirmed on the data. `tt.svd` defaults to `renormalize=True`, which rescales the kept singular values at every split. On `theta.00059` at 4096^2 the stored hybrid fields equal the plain rank-truncated field times one scalar c to $1.2\text{–}3.5 \times 10^{-14}$. The norm is preserved exactly, $E(1)$ shifts by exactly c^2 — $+2.7\%$ at $\rho = 0.014$, falling to $+0.04\%$ at $\rho = 0.64$ — and the gradient flatness ratio is 1.0000, so every scale-invariant statistic is unaffected.

d. The solver must run single-threaded. A step is thousands of small contractions and SVDs, none large enough to amortise thread dispatch. On a 256-core node, four threads instead of one measured the hot einsum in `_apply_core` 43× slower and a whole step 29× slower. The earlier observation that thread count stopped mattering after the move to LAPACK understates it: on many-core hardware extra threads are actively harmful.

e. Forcing at the largest scale does not equilibrate. The approach to a statistically steady state is set by the diffusive time of the forced scale, $1/(\kappa k_f^2)$, which is 200 time units at $k_f = 1$ and was still climbing at $t = 90$ for every amplitude tried. At $k_f = 4$ that time is 12 and the variance plateaus by $t \simeq 40$.

VI. CORRECTIONS TO EARLIER STATEMENTS

Recorded because the reasoning matters more than the conclusions:

- "Translation is rank-preserving" (wrong), then "translation is not rank-preserving, so every step must be lossy" (also wrong, over-corrected). The measured behaviour is the first bullet of the previous section.
- "Apply the seam smoother once per step, inside the loop." It is harmful at every cadence tested.
- "The nonlinear term is the norm-changing one." For a passive scalar with $\nabla\cdot\mathbf{u} = 0$ the transport term conserves variance exactly and the linear diffusive term dissipates; see Eq. (2).
- **The truncation-loss diagnostic was wrong by four orders of magnitude**, and in the alarming direction: it reported numerical dissipation at roughly seventeen times the molecular value when the true figure is about two thousandths of it. It summed every compression, including the dozen or so per right-hand-side evaluation performed on intermediates whose norms are those of the right-hand side rather than of θ. Caught only because it contradicted the variance history, which tracked the reference closely. Replaced by a variance-budget measure, with a warning left on the raw accumulator.
- **Two performance predictions from flop counting, both wrong.** Replacing an einsum with a matmul and cutting the truncation count per step were each expected to help and each measured within noise. The actual win was in a place flop counting does not look at all: the LAPACK backend. Relatedly, the compression granularity inside the transport term was argued in both directions before being settled by measurement.
- **Two wrong diagnoses of a changed variance drift.** A discrepancy against an earlier measurement was attributed first to the timestep and then to the velocity construction; both were ruled out by direct test, and the cause was the grid resolution together with the run length implied by a fixed step count.

a. How the hybrid advantage was quoted. It was first reported as "7.6×, 21×, 90× at $\rho = 0.2$, growing with resolution". That is the error ratio at a fixed budget, a vertical cut through the rate–distortion curve, and it is inflated by the steepness of those curves, increasingly so at larger L. Read as the budget needed for a fixed accuracy the saving is 1.8–2.5 and flat in L. Both describe the same pair of curves; the budget ratio is the one that answers how many parameters the head buys.

b. Reachability of 4096^2. It was described as out of reach. That assumed a fixed compression ratio, which requires $\chi \sim 2^L$. For a fixed *accuracy* the measured requirement is $\chi \sim 2^{0.34L}$, nearly flat ($\chi = 6, 12, 16, 12$ for $L = 7 \dots 10$ at $L_2 = 10^{-2}$), and a step at 4096^2 was measured at 7.3 h for $\chi = 24$ with a peak resident set of 1.4 GB.

VII. STATUS AND NEXT STEPS

a. Done. The validated baseline is committed and separated from the paper material. The cellular-flow cascade answers the central question: the intrinsic rank rises as fine scales are generated and then saturates, flat for twelve time units while the variance decays sevenfold, at ranks well below the geometric bound. Truncation dissipation is negligible against molecular diffusion, so the conditional TDVP upgrade remains unwarranted. Compression was profiled and improved by about an order of magnitude, and the whole campaign now runs as Slurm arrays. The hybrid encoding has been benchmarked against a standard tensor train at four resolutions (companion report `report_vs_standard.pdf`), and the deferred renormalisation check has been settled on the published data.

1. **The DNS-velocity ladder**, at 512^2 with a frozen velocity snapshot. The decisive case: the cellular flow is integrable and its scalar spectrum is steep, $E_\theta \sim k^{-4.9}$, so it is a weak proxy for turbulence, and the measured hybrid advantage drops from 5–24 on that field to 1.2 on random-phase $k^{-5/3}$ fields. *Settles:* what the factor of two becomes on a realistic field, and whether the rank still saturates when the straining is genuinely turbulent.

2. **The forced steady state**, at 256^2 with $f = 0.8 \sin 4y$. *Settles:* whether the rank plateau survives in a sustained cascade rather than a decaying transient.

3. **The production geometry** (4096^2), held rather than blocked. A step costs 7.3 h at $\chi = 24$ and the shared dense-reference cache is written and verified bit-identical; what is missing is a reason to spend the three days, which the two runs above may supply.

4. **2D Navier–Stokes in vorticity form.** Reuses the whole stack apart from the Poisson inversion recovering $\mathbf{u}$ from ω, and makes the transport term genuinely nonlinear, with two quadratic invariants that truncation will damage at different rates.

5. **Conditional: implicit stepping, with variational or sketched products.** The timestep is CFL-limited and an implicit scheme would remove that restriction, at which point AMEn stops being an optimisation of the MPO–MPS product and becomes the enabling solver for the linear system. Sketched products suit the many-small-SVD shape of this workload but replace a certified singular-value tail with a probabilistic bound, which the present methodology relies on. Neither is indicated while the explicit solver with SVD truncation is still producing results.

b. Beyond the passive scalar. The natural extension is 2D Navier–Stokes in vorticity form, $\partial_t \omega + \mathbf{u} \cdot \nabla \omega = \nu \nabla^2 \omega$ with $\mathbf{u}$ recovered from ω by an elliptic solve, which adds a Poisson inversion — a natural fit for the fused/sparse head — and a second quadratic invariant, since 2D conserves both energy and enstrophy and truncation will damage them at different rates.